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SIMATS ENGINEERING
Regno: 192525348

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

81%

8

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
B	Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
C	Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
D	Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
E	Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
F	Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
G	Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
H	Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

40.5

	A	B	C	D	E	F	G	H	
Q1	1.5	2	1	0.5	1	1	0.5	0.5	8
Q2	1	2	0.5	0.5	1	1	0.5	1	8.5
Q3	1.5	1.5	1	1	0.5	1	1	0.5	8
Q4	1	2	0.5	1.5	0.5	0.5	1	1	8
Q5	1.5	1	1	1	1	1	0.8	1	8

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Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Star Topology vs Bus Topology

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and separate cables for each device. Bus topology uses a single backbone cable, so it is cheaper and easier to install.

However, star topology is more reliable because if one device or cable fails, the other devices continue to work. In a bus topology, failure of the main backbone cable can affect the entire network.

Star topology also provides better performance because the central device manages data transmission and reduces collisions. Bus topology may become slower when more devices are connected because all devices share the same communication medium.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Mesh Topology vs Star Topology

Mesh topology provides very high fault tolerance because each node is connected to multiple other nodes. If one connection fails, data can travel through another available path.

Star topology provides moderate fault tolerance. If one individual node or cable fails, the other devices are not affected. However, if the central hub or switch fails, the entire network can stop working.

Therefore, mesh topology is more reliable and suitable for critical environments where continuous communication is important.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Bus Topology vs Mesh Topology

Bus topology is simple and easy to install because it uses a single backbone cable with fewer connections. It is suitable for small networks and requires less cabling.

Mesh topology is more complex because each node must be connected to other nodes. It requires a large number of cables and ports.

As the number of devices increases, the installation of a mesh topology becomes more difficult, time-consuming, and expensive. Therefore, bus topology is easier to install, while mesh topology is more complex.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Hybrid Topology vs Star Topology

Hybrid topology is highly scalable because it combines two or more topologies, such as star, mesh, or bus. New network segments can be added according to organizational requirements without affecting the existing network structure.

Star topology is also scalable, but its expansion depends on the capacity of the central device, such as a switch or hub. When all ports are used, additional hardware is required to connect more devices.

Therefore, hybrid topology provides greater flexibility and scalability than star topology.

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5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Maintenance Comparison of Network Topologies

Star topology is easy to maintain because problems can be quickly identified and isolated to a specific device or cable.

Mesh topology is highly reliable but difficult to maintain because it has many connections. Troubleshooting can be complex and time-consuming.

Bus topology is harder to maintain than star topology because a fault in the main backbone cable can affect the entire network and may be difficult to locate.

Hybrid topology has moderate maintenance complexity. It provides flexibility, but managing different combined topologies requires careful monitoring and skilled administration.

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